EDUARDO CASTRO

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Personal Website (incl. LinkedIn)

EDUCATION

University of Texas Rio Grande Valley — *Edinburg, TX* **August 2023 – December 2027**
Bachelor of Science in Computer Engineering • GPA: 3.44

EXPERIENCE

AI Fellow, Backend Engineering Intern — *RGV AI Hub* **Jul 2026 – Present**

- Developed and extended a Python/FastAPI backend connecting a university website to a locally hosted Qwen LLM through LM Studio and REST APIs.
- Built a Google Sheets-backed RAG pipeline that searches multiple repositories and injects relevant local education, healthcare-program, and career information into model prompts.
- Integrated SQLite for structured local data storage and used Claude Code and Codex alongside manual development to accelerate iteration and shorten production timelines.
- Configured and debugged Linux services, environment variables, Google API credentials, Git workflows, and Cloudflare tunnels for external access.
- Tested multi-user performance and reliability by comparing reasoning settings, token limits, timeout behavior, retries, and response latency for at least four concurrent users.
- Delivered presentations and live demonstrations of the project to professionals and colleagues, explaining technical functionality, project progress, and community applications.

LEADERSHIP & ACTIVITIES

IEEE Student Branch — Project Director **August 2026 – Present**

- Guide student engineering teams through project planning, technical troubleshooting, progress tracking, and workload management.
- IEEE Xtreme Programming Competition: completed a 24-hour international competition solving data structures and algorithms problems under time constraints.

SELECTED ENGINEERING PROJECTS

Computer Vision Object-Following Robot — Team Lead **Nov 2025 – May 2026**

- Led an 8-member team in building a Raspberry Pi autonomous rover that combined embedded hardware, computer vision, and closed-loop movement control.
- Collected and labeled 100+ images using an AI tool, trained a YOLOv8 model to recognize a target object, and converted the model to ONNX for deployment.
- Programmed Python perception and motor-control logic with detection-streak gating to reduce false detections and improve tracking stability.
- Integrated ultrasonic sensors for obstacle and wall avoidance; contributed to circuit design, soldering, Linux/SSH deployment, and system debugging.

Facial Recognition Security System **Jun 2026 – Present**

- Built a Raspberry Pi pipeline for face-image collection, facial encoding, and real-time recognition using Python, OpenCV, and face_recognition.
- Organized labeled user datasets and serialized facial encodings to support repeatable training and recognition across known and unknown visitors.
- Developing a physical locking mechanism and React Native companion app for entry logs and real-time notifications based on camera activity and recognition events.

Adaptive Embedded Control System **May 2026**

- Implemented non-blocking command parsing with a circular RX buffer and manual/autonomous operating modes for sensor-driven outputs.
- Generated PWM with hardware timers, sampled potentiometer input through the ADC, and applied low-pass filtering for autonomous control decisions.
- Added fault-detection states and validated timing and signal behavior with an oscilloscope during hardware testing and debugging.

Club Organizer Web Application **Apr 2026**

- Developed a Django full-stack application with user authentication, event creation, event signup, and a responsive HTML/CSS interface.
- Collaborating through Git/GitHub, Jira, and Agile workflow practices to organize and integrate team contributions.

Focus Timer App **Apr 2026 – Jul 2026**

- Designed and built a Pomodoro app with timers, task management, focus-session tracking, statistics, and user settings.
- Used OpenAI Codex to assist with debugging, code review, enhancements, authentication improvements, and automated testing.
- Implemented secure user authentication and migrated production data storage from SQLite to Neon PostgreSQL for persistent deployment on Render.

TECHNICAL SKILLS

Programming: Python, C++, C (foundational), HTML/CSS, Data Structures & Algorithms
AI / Computer Vision: OpenCV, YOLO model training, face_recognition, RAG, local LLM integration (LM Studio), PyTorch
Backend / Tools: FastAPI, REST APIs, Django, Google Sheets API, Git/GitHub, Linux, SSH, Cloudflare tunnels
Embedded / Hardware: Raspberry Pi, STM32, Arduino, PWM/ADC/timers, motor control, soldering, oscilloscope